

$$z_{xx} = -e^y \sin x$$

T₁₁:

$$u_x = 2x - y \quad u_y = -x + 2y$$

$$u_x(1,1) = 1 \quad u_y(1,1) = 1$$

$$x \frac{\partial u}{\partial x} = u_x(1,1) \cos x + u_y(1,1) \sin x$$

$$\frac{\partial u}{\partial \rho} = \cos x + \sin x = \sqrt{2} \sin(x + \frac{\pi}{4})$$

设

(1) 当 $\alpha = \frac{\pi}{4}$ 时. 在 $(\frac{\pi}{2}, \frac{\pi}{2}) = 1$ 方向上存在函数最大值 $\sqrt{2}$.

(2) $\alpha = -\frac{3}{4}\pi$ 时. 在方向 $1 = (-\frac{\pi}{2}, -\frac{\pi}{2})$ 上函数最小值 $-\sqrt{2}$.

(3) 当 $\alpha = -\frac{\pi}{4}$, $\frac{3}{4}\pi$ 时. 在方向 $1 = (\frac{\pi}{2}, -\frac{\pi}{2})$ 或 $1 = (-\frac{\pi}{2}, \frac{\pi}{2})$ 上函数为 0.

(4) $\text{grad} u = u_x(1,1)i + u_y(1,1)j = i + j$

② T_2 :

$$u_x = yz, \quad u_y = xz, \quad u_z = xy.$$

在 $M(1,1,1)$ 处, $u_x = u_y = u_z = 1$

又向量 I 的方向余弦, $\cos \alpha = \frac{2}{\sqrt{14}}, \cos \beta = -\frac{1}{\sqrt{14}}, \cos \gamma = \frac{3}{\sqrt{14}}$.

~~则 $\frac{\partial u}{\partial l} = u_x(1,1,1)i + u_y(1,1,1)j + u_z(1,1,1)k$~~

$$\begin{aligned} \text{则 } \frac{\partial u}{\partial l} &= u_x(1,1,1)\cos\alpha + u_y(1,1,1)\cos\beta + u_z(1,1,1)\cos\gamma \\ &= \frac{2}{7}\sqrt{14}. \end{aligned}$$

$$\text{grad} u = i + j + k.$$

T_3 : $u_x = 2x + y + z, \quad u_y = 4y + x - 2, \quad u_z = 6z - 6$.

则在 $O(0,0,0)$ 处: $u_x = 3, \quad u_y = -2, \quad u_z = -6$.

$$\text{grad} u = 3i - 2j + 6k, \quad |\text{grad} u| = 7$$

在 $A(1,1,1)$ 点: $u_x=6, u_y=3, u_z=0$.
 可以求得 $grad u = 6i + 3j$ $|grad u| = 3\sqrt{5}$.

T_0 :

(1)
$$\frac{\partial(\alpha u + \beta v)}{\partial x} = \alpha \frac{\partial u}{\partial x} + \beta \frac{\partial v}{\partial x}$$

$$\frac{\partial(\alpha u + \beta v)}{\partial y} = \alpha \frac{\partial u}{\partial y} + \beta \frac{\partial v}{\partial y}$$

则
$$grad(\alpha u + \beta v) = \left[\frac{\partial(\alpha u + \beta v)}{\partial x}, \frac{\partial(\alpha u + \beta v)}{\partial y} \right] = \alpha \left[\frac{\partial u}{\partial x}, \frac{\partial v}{\partial y} \right] + \beta \left[\frac{\partial u}{\partial x}, \frac{\partial v}{\partial y} \right]$$

$$= \alpha grad u + \beta grad v$$

(2)
$$\frac{\partial(u \cdot v)}{\partial x} = v \frac{\partial u}{\partial x} + u \frac{\partial v}{\partial x}$$

(2)
$$\frac{\partial(u \cdot v)}{\partial y} = v \frac{\partial u}{\partial y} + u \frac{\partial v}{\partial y}$$

$$\frac{\partial}{\partial x} \left(\frac{x}{e} + u \frac{\partial u}{\partial y} \right)$$

$$(2) \quad \frac{\partial uv}{\partial x} = v \frac{\partial u}{\partial x} + u \frac{\partial v}{\partial x} \quad \frac{\partial(uv)}{\partial y} = v \frac{\partial u}{\partial y} + u \frac{\partial v}{\partial y}$$

证:

$$\text{grad}(u \cdot v) = \left(\frac{\partial(uv)}{\partial x}, \frac{\partial(uv)}{\partial y} \right) = v \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + u \left(\frac{\partial v}{\partial x}, \frac{\partial v}{\partial y} \right)$$

$$= u \text{grad} v + v \text{grad} u$$

(3)

$$\text{grad} F(u) = \left(\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y} \right) = \left(f'(u) \frac{\partial u}{\partial x}, f'(u) \frac{\partial u}{\partial y} \right)$$

$$= f'(u) \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y} \right) = f'(u) \text{grad} u$$

证:

$$\text{proof: 因 } \frac{\partial r}{\partial x} = \frac{x}{r}, \quad \frac{\partial r}{\partial y} = \frac{y}{r}, \quad \frac{\partial r}{\partial z} = \frac{z}{r}$$

$$\text{则 } \text{grad} \frac{1}{r} = \frac{d}{dr} \left(\frac{1}{r} \right) \text{grad} r = -\frac{1}{r^2} \left(\frac{\partial r}{\partial x} i + \frac{\partial r}{\partial y} j + \frac{\partial r}{\partial z} k \right) = -\frac{1}{r^2} \cdot \frac{1}{r} (xi + yj + zk)$$

$$= -\frac{r}{r^3}$$

§ 4.

$$T_{uv} = e^y \cos x - 2e^x - ye^x$$

T_{11} :

$$f_x(x, y) = 4x - y - 6 \quad f_y(x, y) = -x - 2y - 3$$

$$f_{xx} = 4, \quad f_{xy} = -1, \quad f_{yy} = -2$$

$$f_{xx}^3 = f_{xy} = f_{yy} = 0$$

在点 $(1, -2)$ 处 $f = 5, f_x = 0, f_y = 0, f_{xx} = 4$

$$f_{xy} = -1, \quad f_{yy} = -2$$

$$\text{于是 } f(x, y) = 5 + 2(x-1) - (x-1)(y+2) - (y+2)^2$$

T_2 :

$$f_x = e^x \ln(1+y), \quad f_y = \frac{e^x}{1+y} \quad f_{xx} = e^x \ln(1+y)$$

$$f_{yy} = -\frac{e^x}{(1+y)^2} \quad f_{xy} = \frac{e^x}{1+y}$$

$$f_{xx}^2 = \frac{e^x}{1+y}$$

$$f(x,y) = 5 + 2(x-1) - (x-1)(y+2) - (y+2)^2$$

解:

$$f_x = e^x \ln(1+y), \quad f_y = \frac{e^x}{1+y} \quad f_{xx} = e^x \ln(1+y)$$

$$f_{yy} = -\frac{e^x}{(1+y)^2} \quad f_{xy} = \frac{e^x}{1+y}$$

$$f_{x^3} = e^x \ln(1+y) \quad f_{y^3} = -\frac{2e^x}{(1+y)^3} \quad f_{xy^2} = -\frac{e^x}{(1+y)^2} \quad f_{yx^2} = \frac{e^x}{1+y}$$

则在点 (0,0) 处: $f = 0, \quad f_x = f_x = f_{x^2} = 0$

$$f_y = 1, \quad f_{xy} = 1, \quad f_{yy} = -1, \quad f_{xy^2} = -1, \quad f_{yx^2} = 1, \quad f_{y^3} = -2$$

$$\text{于是 } f(x,y) = y + xy + \frac{1}{2}x^2y - \frac{1}{6}xy^2 + \frac{1}{24}y^3$$

例 5
T1:

(1)

$$\text{令 } z_x = z_y = 0: \quad z_y = -2(y-1) = 0$$

则得 $x=0, y=1$, 于是点 (0,1) 为可能的极值点.
 $B=0, \quad C=-2$, 于是 $H=-4 < 0$, 所以该点为极大值点.

关系, 通过二次函数

$$f_x = e^x \ln(1+y), \quad f_y = \frac{e^x}{1+y} \quad f_{xx} = e^x \ln(1+y),$$

$$f_y = -\frac{e^x}{(1+y)^2} \quad f_{xy} = \frac{e^x}{1+y}$$

$$f_{x^2} = e^x \ln(1+y) \quad f_{xy} = -\frac{2e^x}{(1+y)^2} \quad f_{yy} = -\frac{e^x}{(1+y)^3}$$

$$f_{yx} = \frac{e^x}{1+y} \quad f_{xx} = e^x \ln(1+y) \quad f_{xy} = -\frac{2e^x}{(1+y)^2}$$

$$f_{xx} = f_{xx} = f_{xx} = 0.$$

$$f_y = 1, \quad f_{xy} = 1, \quad f_{yy} = -1, \quad f_{xy} = -1, \quad f_{yx} = 1, \quad f_{yy} = 2.$$

$$f(x,y) = y + xy + \frac{1}{2}x^2y - \frac{1}{3}xy^2 + \frac{1}{3}y^3.$$

§5
T1:

(1)

$$f_x = 2x = 0, \quad f_y = -2(1+y-1) = 0$$

则得 $x=0, y=1$, 于是点 $(0,1)$ 为可微极值点.

$$f_{xx} = 2, \quad f_{xy} = 0, \quad f_{yy} = -2, \text{ 则 } A = 2, \quad B = 0, \quad C = -4 < 0, \text{ 从而此点为极大值.}$$

13)

$$Z = 3axy - x^2 - y^2 \quad (a > 0).$$

$$\text{令 } Z_x = 3ay - 2x = 0, \quad Z_y = 3ax - 2y = 0.$$

$$\text{则 } \begin{cases} x_1 = 0 \\ y_1 = 0 \end{cases} \quad \begin{cases} x_2 = a \\ y_2 = a \end{cases}$$

即 在点 $(0,0)$, (a,a) 处可以有极值.

$$Z_{xx} = -2, \quad Z_{xy} = 3a, \quad Z_{yy} = -2.$$

则在点 $(0,0)$, $A = 0, B = 3a, C = 0$. 于是 $H = -9a^2 < 0$. 此时函数无极值.

在点 (a,a) , $A = -2, B = 3a, C = -2$. 于是 $H = 27a^2 > 0$. 此时函数有极大值 $Z = a^3$.

(15)

$$Z = y \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}.$$

$$\frac{\partial Z}{\partial x} = y \cdot \frac{-\frac{2x}{a^2}}{2\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}} = 0 \quad \Rightarrow \quad x = 0.$$

$$\frac{\partial Z}{\partial y} = \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}} - \frac{y^2}{b^2 \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}} = 0 \quad \Rightarrow \quad y = 0.$$

在点 (a, a) . $A = -6a < 0$. $B = 3a$. $C = -6a$, 于是 $H = 27a^3 > 0$. 此时函数有极大值 $z = a^3$

(5)

$$L_2 z_x = y \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}} - \frac{x^2 y}{a^2 \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}} = 0$$

$$z_y = x \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}} - \frac{x y^2}{b^2 \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}} = 0$$

则得:
$$\begin{cases} x_1 = 0 \\ y_1 = 0 \end{cases} \quad \begin{cases} x_2 = \frac{a}{\sqrt{3}} \\ y_2 = \frac{b}{\sqrt{3}} \end{cases} \quad \begin{cases} x_3 = -\frac{a}{\sqrt{3}} \\ y_3 = -\frac{b}{\sqrt{3}} \end{cases} \quad \begin{cases} x_4 = \frac{a}{\sqrt{3}} \\ y_4 = -\frac{b}{\sqrt{3}} \end{cases} \quad \begin{cases} x_5 = -\frac{a}{\sqrt{3}} \\ y_5 = \frac{b}{\sqrt{3}} \end{cases}$$

于是我们在点 $P_1(0,0)$, $P_2(\frac{a}{\sqrt{3}}, \frac{b}{\sqrt{3}})$, $P_3(-\frac{a}{\sqrt{3}}, -\frac{b}{\sqrt{3}})$, $P_4(\frac{a}{\sqrt{3}}, -\frac{b}{\sqrt{3}})$, $P_5(-\frac{a}{\sqrt{3}}, \frac{b}{\sqrt{3}})$ 处可能取极值

$P_5(-\frac{a}{\sqrt{3}}, \frac{b}{\sqrt{3}})$ 处可能取极值

在 $P_2(\frac{a}{\sqrt{3}}, \frac{b}{\sqrt{3}})$, $P_3(-\frac{a}{\sqrt{3}}, -\frac{b}{\sqrt{3}})$ 处, $A = -\frac{4\sqrt{3}b}{3a} < 0$, $B = -\frac{2}{3}\sqrt{3}$, $C = -\frac{4\sqrt{3}a}{3b}$

此时 $H = 4 > 0$. 于是函数有极大值 $z = \frac{\sqrt{3}ab}{9}$.

在 $P_4(\frac{a}{\sqrt{3}}, -\frac{b}{\sqrt{3}})$, $P_5(-\frac{a}{\sqrt{3}}, \frac{b}{\sqrt{3}})$ 处, $A = \frac{4\sqrt{3}b}{3a} > 0$, $B = -\frac{2}{3}\sqrt{3}$, $C = -\frac{4\sqrt{3}a}{3b}$.

它是 a 和 b 的函数
 $s(a, b)$ 达到最小

这里为什么
 有正有负, 如果
 和很小, 却不
 可以保证
 就为了选

$$Zg = x \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}} - \frac{xy}{c}$$

$$L^2 \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}} - \frac{xy}{c}$$

则得:
$$\begin{cases} x_1 = 0 \\ y_1 = 0 \end{cases} \quad \begin{cases} x_2 = \frac{a}{\sqrt{3}} \\ y_2 = \frac{b}{\sqrt{3}} \end{cases} \quad \begin{cases} x_3 = -\frac{a}{\sqrt{3}} \\ y_3 = -\frac{b}{\sqrt{3}} \end{cases} \quad \begin{cases} x_4 = \frac{a}{\sqrt{3}} \\ y_4 = -\frac{b}{\sqrt{3}} \end{cases} \quad \begin{cases} x_5 = -\frac{a}{\sqrt{3}} \\ y_5 = \frac{b}{\sqrt{3}} \end{cases}$$

于是我们在点 $P_1(0,0)$, $P_2(\frac{a}{\sqrt{3}}, \frac{b}{\sqrt{3}})$

$P_3(-\frac{a}{\sqrt{3}}, \frac{b}{\sqrt{3}})$ 处可取极值

在 $P_2(\frac{a}{\sqrt{3}}, \frac{b}{\sqrt{3}})$, $P_3(-\frac{a}{\sqrt{3}}, -\frac{b}{\sqrt{3}})$ 处, $A = -\frac{4\sqrt{3}b}{3a} < 0$ $B = -\frac{2}{3}\sqrt{3}$ $C = -\frac{4\sqrt{3}a}{3b}$

此时 $H=4>0$, 于是函数有极大值 $Z = \frac{4}{9}ab$

在 $P_4(\frac{a}{\sqrt{3}}, -\frac{b}{\sqrt{3}})$, $P_5(-\frac{a}{\sqrt{3}}, \frac{b}{\sqrt{3}})$ 处, $A = \frac{4\sqrt{3}b}{3a} > 0$, $B = -\frac{2}{3}\sqrt{3}$ $C = \frac{4\sqrt{3}a}{3b}$

此时 $H=4>0$, 于是函数有极小值 $Z = -\frac{4}{9}ab$

反看可

前进方向. 代表中国广大人民的根本利益.

在 $(0,0)$ 处, $A=0, B=1, C=0$. 此时 $H=-1 < 0$, 于是函数无极值.

T_2 : Proof: $\frac{\partial z}{\partial x} = -(1+e^y) \sin x = 0, \frac{\partial z}{\partial y} = e^y \cos x - e^y - ye^y \stackrel{=0}{\neq 0}$.

因 $1+e^y \neq 0$, 则 $\sin x = 0, z_y = e^y \cos x - e^y - ye^y = 0$.

又 $e^y \neq 0$, 则 $\cos x - 1 = y = 0$. 即有 $y = \cos x - 1$.

当 k 为偶数时, $y=0$; 当 k 为奇数时, $y=-2$. 可能的极值点为 $\begin{cases} x_1 = 2k\pi \\ x_2 = \end{cases} \begin{cases} y_1 = 0 \\ y_2 = \end{cases}$

$$Z_x^2 = -(1+e^y) \cos x \quad Z_{xy} = -e^y \sin x \quad Z_y^2 = e^y \cos x - 2e^y - ye^y$$

则在点 $(2k\pi, 0)$, $A = -2 < 0$, $B = 0$, $C = -1$, 此时 $H = 0$.

则此时函数有最大值 $z = 2$.

在点 $((2k+1)\pi, -2)$, $A = 1 + \frac{1}{e^y}$, $B = 0$, $C = -\frac{1}{e^y}$, 此时 $H = -\frac{1}{e^y} (1 + \frac{1}{e^y}) < 0$.

则此时函数无极值.

综上可知, 函数 $z = (1+e^y) \cos x - y \cdot e^y$ 有无穷多个极大值, 但无极小值.

T₃:

设三角形边长为 x, y, z , $C = x+y+z = 2p$, $D = \frac{x+y+z}{2} = p$. 于是 $z = 2p - x - y$.

$$\text{则 } S = \sqrt{D(D-x)(D-y)(D-z)} = \sqrt{p(p-x)(p-y)(x+y-p)}.$$

$$\text{考虑函数 } u = S^2 = p(p-x)(p-y)(x+y-p), \quad 0 < x, y < p$$

C 的取值均为 u 的极值且当 u 在点 (x, y) 取得的极值不为 0 时,

考虑函数 $u = S^2 = p(p-x)(p-y)(x+y-p)$, $0 < x, y < p$

S 的极值均为 u 的极值且当 u 在点 (x, y) 取得的极值不为 0 时,

S 也在点 (x, y) 取得极值.

令 $u_x = p(p-y)(p-x-y) = 0$, $u_y = p(p-x)(p-x-y) = 0$.

因 $p \neq 0$, $0 < x, y < p$, $\Rightarrow x=y = \frac{2}{3}p$. 于是 $z = \frac{2}{3}p$.

故当 $x=y=z = \frac{2}{3}p$ 时, 面积最大且值为 $S = \frac{\sqrt{3}}{4}p^2$.

14:

$$\begin{cases} z_x = x - 4y + 3 = 0 \\ z_y = -4x + 18y - 14 = 0 \end{cases}$$

解得: $\begin{cases} x=1 \\ y=1 \end{cases}$ 即在点 $(1, 1)$ 可能取得极值

$z_{xx} = 1$, $z_{xy} = -4$, $z_{yy} = 18$, 则 $A = 1 > 0$, $B = -4$, $C = 18$. 于是 $H = 22 > 0$.

则此时函数有最小值 $z = -5$, 双叶曲面有最低点 $(1, 1, -5)$

又当 $x^2 + y^2 \rightarrow +\infty$ 时, $z \rightarrow +\infty$, 故曲面无最高点.

来反
上看
b 可
t₁